

Problem 1

Let $S(n, k)$ denote the number of ways to partition $\{1, 2, \dots, n\}$ into k nonempty, unlabeled subsets. Give a story proof for each of the following identities.

(a)

Prove that

$$S(n+1, k) = S(n, k-1) + kS(n, k).$$

solution:

Consider partitioning the set

$$\{1, 2, \dots, n, n+1\}$$

into k nonempty, unlabeled subsets.

Case 1: The element $n+1$ forms a subset by itself.

Then the remaining n elements

$$\{1, 2, \dots, n\}$$

must be partitioned into $k-1$ nonempty subsets. There are

$$S(n, k-1)$$

ways.

Case 2: The element $n+1$ does not form a subset by itself.

First partition

$$\{1, 2, \dots, n\}$$

into k nonempty subsets. This can be done in

$$S(n, k)$$

ways.

Then $n+1$ can be placed into any one of these k subsets, giving

$$kS(n, k)$$

possibilities.

the two cases are disjoint and cover all possible partitions

$$\implies S(n+1, k) = S(n, k-1) + kS(n, k)$$

(b)

Prove that

$$S(n, k) = \sum_{j=k-1}^{n-1} \binom{n-1}{j} S(j, k-1).$$

solution:

Consider partitioning

$$\{1, 2, \dots, n\}$$

into k nonempty, unlabeled subsets.

Focus on the subset containing the element n .

Let j denote the number of elements that are not in the same subset as n . These j elements must form the other $k-1$ subsets.

First, choose these j elements from

$$\{1, 2, \dots, n-1\}.$$

There are

$$\binom{n-1}{j}$$

ways to choose them. $\therefore$ partition these j chosen elements into $k-1$ nonempty subsets.

$$\therefore S(j, k-1)$$

ways.

The remaining $n - 1 - j$ elements must all belong to the same subset as n , so there are no further choices. Therefore, for a fixed value of j , the number of partitions is

$$\binom{n-1}{j} S(j, k-1).$$

the j elements must form $k - 1$ nonempty subsets

$$\implies j \geq k - 1.$$

$$j \leq n - 1.$$

$$\implies S(n, k) = \sum_{j=k-1}^{n-1} \binom{n-1}{j} S(j, k-1)$$

Problem 2

A nonrepeat password is a nonempty sequence formed from m distinct symbols, where no symbol may appear more than once. The length of the password is ℓ , which could be any integer from 1 to m . Suppose that one password is chosen uniformly at random from all nonrepeat passwords. Let p_m be the probability that the selected password uses all m symbols.

(a)

(a) Find an expression for p_m .

solution

For a password of length ℓ , the number of the ℓ -length password is

$$\binom{m}{\ell} * \ell! = \frac{m!}{(m-\ell)!}$$

since the total number of nonrepeat passwords is

$$\sum_{\ell=1}^m \frac{m!}{(m-\ell)!}.$$

The number of m -length password is

$$m!$$

$$\therefore p_m = \frac{m!}{\sum_{\ell=1}^m \frac{m!}{(m-\ell)!}}$$

(b)

(b) Show that

$$p_m = \frac{1}{m \sum_{j=0}^{m-1} \frac{1}{j!}},$$

and

$$\lim_{m \rightarrow \infty} p_m = \frac{1}{e},$$

where e is the base of the natural logarithm.

Solution.

$$\begin{aligned} p_m &= \frac{m!}{\sum_{\ell=1}^m \frac{m!}{(m-\ell)!}}. \\ \implies p_m &= \frac{m!}{m! \sum_{\ell=1}^m \frac{1}{(m-\ell)!}}. \end{aligned}$$

$$\Rightarrow p_m = \frac{1}{\sum_{\ell=1}^m \frac{1}{(m-\ell)!}}.$$

Let

$$j = m - \ell.$$

Thus

$$j \in [0, m-1]$$

$$\sum_{\ell=1}^m \frac{1}{(m-\ell)!} = \sum_{j=0}^{m-1} \frac{1}{j!}.$$

$$\Rightarrow p_m = \frac{1}{\sum_{j=0}^{m-1} \frac{1}{j!}}$$

From Taylor series expansion :

$$e^x = \sum_{j=0}^{\infty} \frac{x^j}{j!}.$$

When $x = 1$:

$$e = \sum_{j=0}^{\infty} \frac{1}{j!}.$$

$$\Rightarrow \lim_{m \rightarrow \infty} \sum_{j=0}^{m-1} \frac{1}{j!} = e.$$

Thus

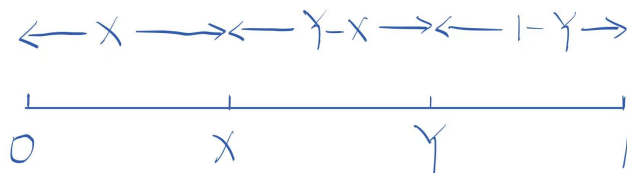
$$\boxed{\lim_{m \rightarrow \infty} p_m = \frac{1}{e}}.$$

Problem 3

(Geometry Probability) You get a stick and break it randomly into three pieces. Two breaking points are chosen independently and uniformly. What is the probability that you can make a triangle using such three pieces?

Solution: assume there are two breaking point, the coordinates are X and Y,

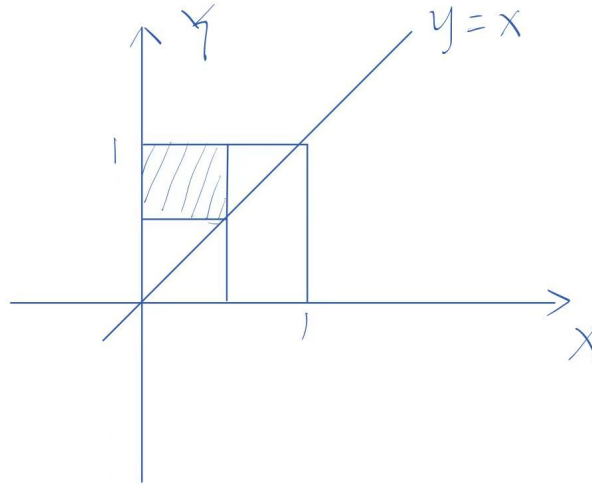
$$0 < X < Y < 1.$$



$$\begin{cases} X + (Y - X) \geq (1 - Y) \\ X + (1 - Y) \geq (Y - X) \\ (Y - X) + (1 - Y) \geq X \end{cases}$$

$$\implies \begin{cases} X \leq \frac{1}{2} \\ Y \geq \frac{1}{2} \\ Y \geq X \end{cases}$$

the total area is the square, the region satisfying the requirements is the shaded area.



$$\therefore A_{\text{total}} = 1 * 1 = 1.$$

$$A_{\text{shaded}} = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

$$\implies P = \frac{A_{\text{shaded}}}{A_{\text{total}}} = \frac{\frac{1}{4}}{1} = \frac{1}{4}.$$

Problem 4

Problem 4 (20%)

(Birthday Problem) In the birthday problem, we assumed that all 365 days of the year are equally likely (and excluded February 29). In reality, some days are slightly more likely as birthdays than others. For example, scientists have long struggled to understand why more babies are born 9 months after a holiday.

Let

$$\mathbf{p} = (p_1, p_2, \dots, p_{365})$$

be the vector of birthday probabilities, with p_j the probability of being born on the j th day of the year (February 29 is still excluded, with no offense intended to Leap Dayers). The k th elementary symmetric polynomial in the variables $x_1, \dots, x_n$ is defined by

$$e_k(x_1, \dots, x_n) = \sum_{1 \leq j_1 < j_2 < \dots < j_k \leq n} x_{j_1} \cdots x_{j_k}, \quad 2 \leq k \leq 365.$$

This just says to add up all of the

$$\binom{n}{k}$$

terms we can get by choosing and multiplying k of the variables.

For example,

$$\begin{aligned}e_1(x_1, x_2, x_3) &= x_1 + x_2 + x_3, \\e_2(x_1, x_2, x_3) &= x_1x_2 + x_1x_3 + x_2x_3,\end{aligned}$$

and

$$e_3(x_1, x_2, x_3) = x_1x_2x_3.$$

Now let $k \geq 2$ be the number of people.

- (a) Find a simple expression for the probability that there is at least one birthday match, in terms of $\mathbf{p}$ and an elementary symmetric polynomial.
- (b) Explain intuitively why it makes sense that

$$P(\text{at least one birthday match})$$

is minimized when

$$p_j = \frac{1}{365}$$

for all j , by considering simple and extreme cases.

- (c) The famous arithmetic mean–geometric mean inequality says that for $x, y \geq 0$,

$$\frac{x+y}{2} \geq \sqrt{xy}.$$

This inequality follows from adding $4xy$ to both sides of

$$x^2 - 2xy + y^2 = (x - y)^2 \geq 0.$$

Define

$$\mathbf{r} = (r_1, \dots, r_{365})$$

by

$$r_1 = r_2 = \frac{p_1 + p_2}{2}, \quad r_j = p_j \quad \text{for } 3 \leq j \leq 365.$$

Using the arithmetic mean–geometric mean bound and the fact, which you should verify, that

$$\begin{aligned}e_k(x_1, \dots, x_n) &= x_1x_2e_{k-2}(x_3, \dots, x_n) \\&\quad + (x_1 + x_2)e_{k-1}(x_3, \dots, x_n) \\&\quad + e_k(x_3, \dots, x_n),\end{aligned}$$

show that

$$P(\text{at least one birthday match} \mid \mathbf{p}) \geq P(\text{at least one birthday match} \mid \mathbf{r})$$

with strict inequality if $\mathbf{p} \neq \mathbf{r}$, where the given $\mathbf{r}$ notation means that the birthday probabilities are given by $\mathbf{r}$.

Using this, show that the value of $\mathbf{p}$ that minimizes the probability of at least one birthday match is given by

$$p_j = \frac{1}{365}$$

for all j .

Solution:

(a)

Assume the event that at least one birthday match is A

The probability that all k birthdays are distinct is B

$$P(\text{B} \mid \mathbf{p}) = k! \sum_{1 \leq j_1 < \dots < j_k \leq 365} p_{j_1} \cdots p_{j_k}.$$

$$\therefore P(\text{all distinct} \mid \mathbf{p}) = k!e_k(p_1, \dots, p_{365}).$$

$$\therefore P(\text{at least one match} \mid \mathbf{p}) = 1 - k!e_k(p_1, \dots, p_{365})$$

(b)

For $k = 2$,

$$P(\text{match}) = \sum_{j=1}^{365} p_j^2.$$

If

$$p_j = \frac{1}{365},$$

then

$$P(\text{match}) = 365 \left(\frac{1}{365} \right)^2 = \frac{1}{365}.$$

In an extreme case,

$$p_1 = 1, \quad p_2 = \cdots = p_{365} = 0,$$

so for $k \geq 2$,

$$P(\text{at least one match}) = 1.$$

Thus, concentrating probability on fewer days increases collisions, while

$$p_1 = p_2 = \cdots = p_{365} = \frac{1}{365}$$

spreads birthdays most evenly and minimizes the probability of a match.

(c)

First,

$$e_k(x_1, \dots, x_n) = x_1 x_2 e_{k-2}(x_3, \dots, x_n) + (x_1 + x_2) e_{k-1}(x_3, \dots, x_n) + e_k(x_3, \dots, x_n).$$

Define

$$r_1 = r_2 = \frac{p_1 + p_2}{2}, \quad r_j = p_j, \quad 3 \leq j \leq 365.$$

Then

$$r_1 + r_2 = p_1 + p_2.$$

Also, by AM-GM,

$$\frac{p_1 + p_2}{2} \geq \sqrt{p_1 p_2},$$

so

$$r_1 r_2 = \left(\frac{p_1 + p_2}{2} \right)^2 \geq p_1 p_2.$$

Hence

$$\begin{aligned} e_k(\mathbf{r}) - e_k(\mathbf{p}) &= (r_1 r_2 - p_1 p_2) e_{k-2}(p_3, \dots, p_{365}) \\ &= \left[\left(\frac{p_1 + p_2}{2} \right)^2 - p_1 p_2 \right] e_{k-2}(p_3, \dots, p_{365}) \\ &= \frac{(p_1 - p_2)^2}{4} e_{k-2}(p_3, \dots, p_{365}) \\ &\geq 0. \end{aligned}$$

Therefore,

$$e_k(\mathbf{r}) \geq e_k(\mathbf{p}).$$

Using part (a),

$$P(\text{match} \mid \mathbf{p}) = 1 - k!e_k(\mathbf{p}),$$

and

$$P(\text{match} \mid \mathbf{r}) = 1 - k!e_k(\mathbf{r}).$$

Thus,

$$P(\text{match} \mid \mathbf{p}) \geq P(\text{match} \mid \mathbf{r})$$

If $p_1 \neq p_2$, then

$$(p_1 - p_2)^2 > 0,$$

and hence, when $e_{k-2} > 0$,

$$P(\text{match} \mid \mathbf{p}) > P(\text{match} \mid \mathbf{r}).$$

Thus averaging two unequal probabilities decreases the match probability. Repeated averaging gives

$$p_1 = p_2 = \cdots = p_{365}.$$

Since

$$\sum_{j=1}^{365} p_j = 1,$$

we obtain

$$365p_j = 1,$$

so

$$p_j = \frac{1}{365}, \quad j = 1, \dots, 365$$

Therefore,

$$P(\text{at least one birthday match}) \text{ is minimized at } p_j = \frac{1}{365}$$

Problem 5

Problem 5 (20%)

A card collection contains five different types of cards. Each independent draw produces card type i with probability p_i , where

$$(p_1, p_2, p_3, p_4, p_5) = (0.05, 0.15, 0.20, 0.25, 0.35).$$

Let Q_n denote the probability of collecting all five card types after n independent draws, $n \geq 1$.

For each $i \in \{1, \dots, 5\}$, define

$$E_i = \{\text{card type } i \text{ does not appear in the first } n \text{ draws}\}.$$

For any index set

$$I \subseteq \{1, \dots, 5\},$$

define

$$E_I = \bigcap_{i \in I} E_i.$$

Thus, E_I is the event that none of the card types whose indices belong to I appears in the first n draws. By convention,

$$E_\emptyset = \Omega,$$

where Ω denotes the entire sample space.

(a) Use the inclusion-exclusion principle to prove that

$$Q_n = \sum_{I \subseteq \{1, \dots, 5\}} (-1)^{|I|} \left(1 - \sum_{i \in I} p_i\right)^n,$$

where $|I|$ denotes the number of elements in I .

(b) Find the smallest integer n such that

$$Q_n \geq 0.90.$$

You can write some code to calculate it. Include the screenshot of your code and output in the terminal which can support your answer.

(c) Find the smallest n for which the probability of collecting all five types is at least 0.90 in the uniform case. (No need to include the code.) Write code to plot Q_n as a function of n for both equal and unequal p_i , along with their corresponding probabilities. (You should include the graph.) Compare Q_n with the corresponding probability between the two cases.

Solution:

(a)

Collecting all five card types means that none of the events $E_1, \dots, E_5$ occurs. Hence

$$Q_n = 1 - P\left(\bigcup_{i=1}^5 E_i\right).$$

By the inclusion-exclusion principle,

$$Q_n = 1 - \sum_i P(E_i) + \sum_{i < j} P(E_i \cap E_j) - \sum_{i < j < k} P(E_i \cap E_j \cap E_k) + \dots.$$

Based on the question stem

$$E_I = \bigcap_{i \in I} E_i$$

For one draw,

$$P(\text{no type in } I) = 1 - \sum_{i \in I} p_i.$$

the n draws are independent,

$$P(E_I) = \left(1 - \sum_{i \in I} p_i\right)^n.$$

$$Q_n = \sum_{I \subseteq \{1, \dots, 5\}} (-1)^{|I|} P(E_I).$$

$$\Rightarrow Q_n = \sum_{I \subseteq \{1, \dots, 5\}} (-1)^{|I|} \left(1 - \sum_{i \in I} p_i\right)^n$$

For $I = \emptyset$,

$$E_{\emptyset} = \Omega, \quad P(E_{\emptyset}) = 1,$$

so the term corresponding to $I = \emptyset$ gives the initial 1 in the inclusion-exclusion formula.

(b)

Using

$$(p_1, p_2, p_3, p_4, p_5) = (0.05, 0.15, 0.20, 0.25, 0.35),$$

we compute

$$Q_n = \sum_{I \subseteq \{1, \dots, 5\}} (-1)^{|I|} \left(1 - \sum_{i \in I} p_i\right)^n.$$

A numerical search gives

$$Q_{45} = 0.8998932711 < 0.90,$$

while

$$Q_{46} = 0.9049652386 > 0.90.$$

Hence the smallest integer satisfying

$$Q_n \geq 0.90$$

is

$$n = 46$$

The code and terminal output are shown below.

The screenshot shows a code editor with a Python script named `code1.py` and a terminal window below it. The code defines a function `Q(n)` that calculates a value based on a list `p` and iterates over combinations of indices. The terminal output shows the results of running the script for `n=45` and `n=46`.

```

1 import itertools
2
3 p = [0.05, 0.15, 0.20, 0.25, 0.35]
4
5 def Q(n):
6     total = 0.0
7
8     for r in range(6):
9         for I in itertools.combinations(range(5), r):
10
11             s = sum(p[i] for i in I)
12
13             total += (-1)**r * (1 - s)**n
14
15     return total
16
17 n = 1
18
19 while Q(n) < 0.90:
20     n += 1
21     print("smallest n =", n)
22     print("Q_{ } = {:.10f}".format(n - 1, Q(n - 1)))
23     print("Q_{ } = {:.10f}".format(n, Q(n)))

```

```

jasmin@localhost:~/latex$ ~/.venvs/probability/bin/python -u "/home/jasmin/latex/SI140/code1.py"
Q_46 = 0.9049652386
jasmin@localhost:~/latex$ ~/.venvs/probability/bin/python -u "/home/jasmin/latex/SI140/code1.py"
smallest n = 46
Q_45 = 0.8998932711
Q_46 = 0.9049652386
jasmin@localhost:~/latex$

```

Figure 1: Code and terminal output for part (b).

(c)

$$p_1 = p_2 = p_3 = p_4 = p_5 = \frac{1}{5}.$$

If $|I| = k$

$$\implies \sum_{i \in I} p_i = \frac{k}{5}.$$

$$\begin{aligned} \implies Q_n &= \sum_{k=0}^5 (-1)^k \binom{5}{k} \left(1 - \frac{k}{5}\right)^n. \\ &= 1 - 5 \left(\frac{4}{5}\right)^n + 10 \left(\frac{3}{5}\right)^n - 10 \left(\frac{2}{5}\right)^n + 5 \left(\frac{1}{5}\right)^n. \end{aligned}$$

notice that

$$Q_{17} = 0.8891009573 < 0.90,$$

$$Q_{18} = 0.9109429198 > 0.90.$$

the smallest integer in the uniform case is

$$n = 18$$

For comparison,

$$n_{\text{uniform}} = 18, \quad n_{\text{unequal}} = 46.$$

The probabilities Q_n for the two cases are plotted below.

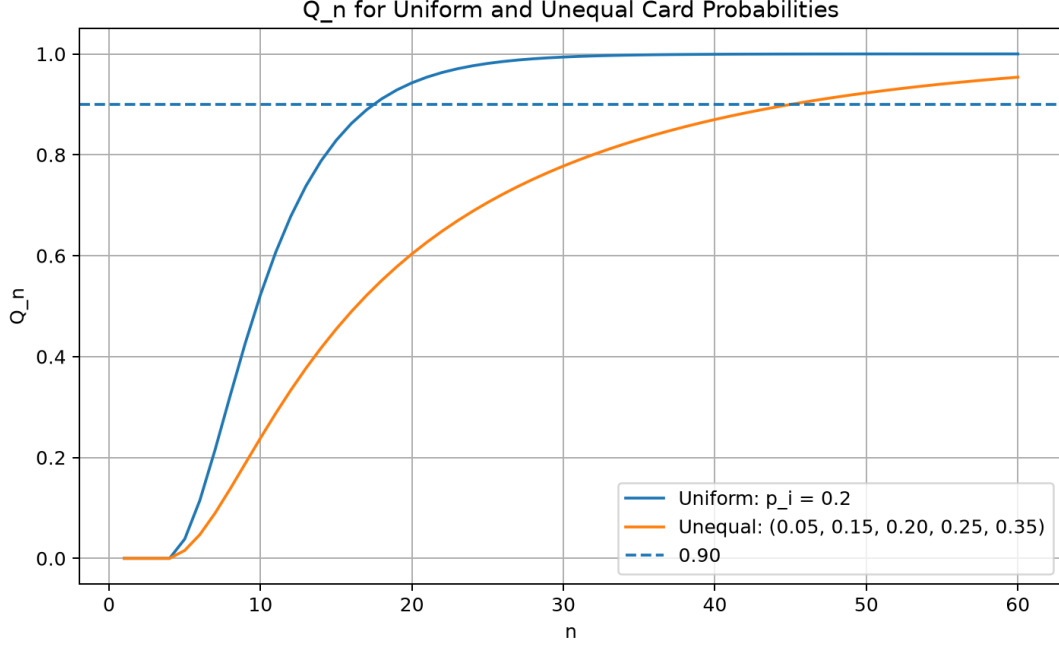


Figure 2: Q_n for uniform and unequal card probabilities.

From the graph,

$$Q_n^{\text{uniform}} > Q_n^{\text{unequal}}$$

for the relevant values of n .

In the unequal case, the rarest card has probability

$$p_1 = 0.05,$$

so it is much more difficult to collect all five card types. In the uniform case, every type has probability

$$p_i = 0.20,$$

so no single card type becomes a bottleneck. Therefore, the probability of collecting all five types increases much faster in the uniform case.